Sustainability in the AI Policies of Open Source Projects: A Grey Literature Review

Idea note for ICT4S 2027

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Motivation

- Open source software (OSS) projects increasingly rely on generative AI (GenAI): coding agents were adopted by 22–29% of 128,018 GitHub projects within months of their release (Robbes et al. 2026).
- Practices are highly diverse: no convention exists, even on what contributors must disclose, and many policy files are revised within months (Hora et al. 2026).
- Policies are written mostly in reaction to GenAI ("AI slop", review load, licensing), not out of a sustainability agenda (Hora et al. 2026).
- Sustainability is a *wicked problem* (Rittel and Webber 1973): its dimensions are interdependent, improving one can degrade another, and no single objective can be optimised.
- We do not know whether, and how, OSS communities frame sustainability when they regulate GenAI.

Research questions and hypotheses

- **RQ1.** Is sustainability addressed in the stated purpose (*raison d'être*) of open source projects?
  - *H1:* yes, or partially.
- **RQ2.** Is sustainability addressed in the AI policies of open source projects?
  - *H2:* no, or only partially.
- **RQ3.** Which definitions and characteristics of sustainability are addressed in these AI policies?
  - *H3:* mostly technical and individual sustainability, centred on the maintainer (workload, code quality, burnout).
- **RQ4.** Do these definitions provide frameworks leading to strong sustainability?
  - *H4:* no.

Methodology

We follow the guidelines for multivocal and grey literature reviews of Garousi et al. (2019). AI policies are grey literature (`CONTRIBUTING.md`, `AI_POLICY.md`, governance pages, discussion threads), so a classical systematic literature review does not apply.

Planning

- Define the review goal and the four RQs above.
- Decide that grey literature is the primary source, because academic work does not yet cover policies produced by communities.
- Complement it with a short academic search to ground the sustainability framework used in the coding scheme.

Source selection and search

- **Project sample.** Select OSS projects by explicit criteria (for example popularity, foundation membership such as Linux Foundation or Apache, domain diversity), using the corpus of Hora et al. (2026) as a starting point.
- **Policy files.** Collect AI policies from the repositories, project websites, and governance documents.
- **Purpose statements (RQ1).** Collect README, mission statements, manifestos, and charters.
- **Search.** Keyword queries (e.g., "AI policy", "LLM", "Copilot", "generated code") on GitHub code search and web search engines, with snowballing from links and mentions.
- **Stopping criterion.** Stop when new sources no longer add new codes (theoretical saturation).
- **Freezing.** Fix a collection date and archive every document (Software Heritage, Wayback Machine) for reproducibility, since policies change quickly.

Inclusion and quality assessment

- **Inclusion.** The document states rules or positions on AI-generated contributions or AI tool use in the project.
- **Exclusion.** Duplicates, forks that copy a policy verbatim (counted once), documents not in English or French, and unofficial opinions of individuals.
- **Quality.** Apply the grey literature criteria of Garousi et al. (2019): authority of the producer, methodology, objectivity, date, novelty, and impact.
- Record whether each document is official (merged policy) or a proposal under discussion.

Data extraction

- Project metadata: size, age, governance, foundation, domain.
- Policy metadata: date, version history, stance (ban, permit with conditions, permit freely), disclosure requirement.
- Verbatim excerpts that mention sustainability or related terms.

Coding scheme and synthesis

- **Sustainability dimensions.** Deductive codes from the Karlskrona manifesto (Becker et al. 2015): individual, social, economic, environmental, and technical. Inductive codes are added as they emerge.
- **Strength of sustainability.** Each definition is classified as *weak* (substitutability between natural and other capital) or *strong* (critical natural capital must be preserved) (Neumayer 2013).

- **Procedure.** Thematic analysis (Braun and Clarke 2006), with two coders on a sample, inter-rater agreement (Cohen's kappa), and disagreements resolved by discussion.
- **Mapping to RQs.**
  - RQ1: presence or absence of sustainability in purpose statements.
  - RQ2: presence or absence of sustainability in AI policies.
  - RQ3: distribution of dimensions and definitions across policies.
  - RQ4: classification of definitions as weak or strong, and analysis of interdependencies between dimensions.

Sustainability coding taxonomy

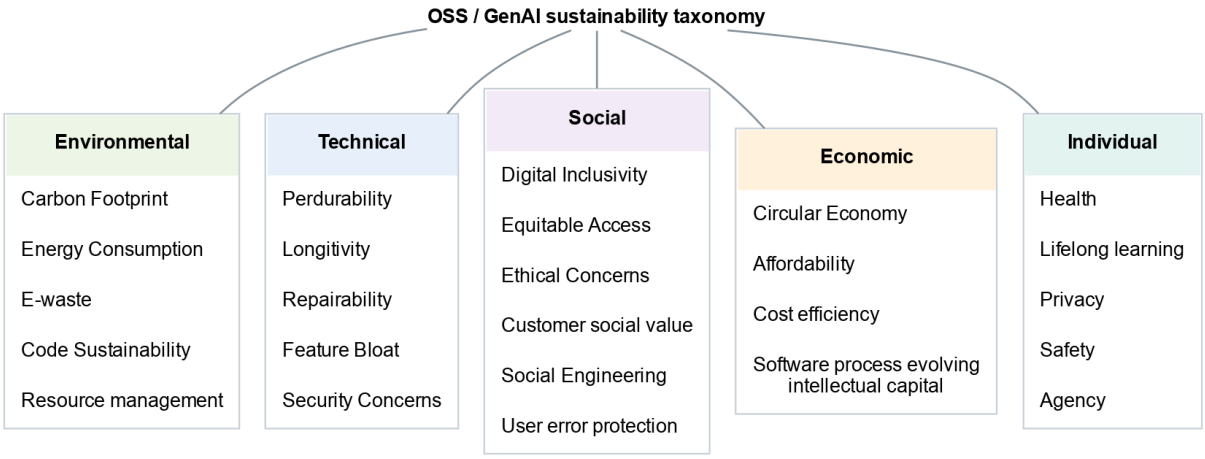
The five high-level dimensions follow the Karlskrona manifesto (Becker et al. 2015). The twenty categories assigned to the environmental, technical, social and economic dimensions are drawn from Roy et al. (Roy et al. 2026). The five individual-level categories are provisional literature-derived extensions whose original definitions require verification. Operational definitions and coding guidance are adaptations by the research team for OSS/GenAI.

Dimension	Category	Provenance
Environmental	Carbon Footprint	Roy et al. (Roy et al. 2026)
Environmental	Energy Consumption	Roy et al. (Roy et al. 2026)
Environmental	E-waste	Roy et al. (Roy et al. 2026)
Environmental	Code Sustainability	Roy et al. (Roy et al. 2026)
Environmental	Resource management	Roy et al. (Roy et al. 2026)
Technical	Perdurability	Roy et al. (Roy et al. 2026)
Technical	Longitivity	Roy et al. (Roy et al. 2026)
Technical	Repairability	Roy et al. (Roy et al. 2026)
Technical	Feature Bloat	Roy et al. (Roy et al. 2026)
Technical	Security Concerns	Roy et al. (Roy et al. 2026)
Social	Digital Inclusivity	Roy et al. (Roy et al. 2026)
Social	Equitable Access	Roy et al. (Roy et al. 2026)
Social	Ethical Concerns	Roy et al. (Roy et al. 2026)
Social	Customer social value	Roy et al. (Roy et al. 2026)
Social	Social Engineering	Roy et al. (Roy et al. 2026)
Social	User error protection	Roy et al. (Roy et al. 2026)
Economic	Circular Economy	Roy et al. (Roy et al. 2026)
Economic	Affordability	Roy et al. (Roy et al. 2026)
Economic	Cost efficiency	Roy et al. (Roy et al. 2026)
Economic	Software process evolving intellectual capital	Roy et al. (Roy et al. 2026)
Individual	Health	Provisional literature extension
Individual	Lifelong learning	Provisional literature extension
Individual	Privacy	Provisional literature extension
Individual	Safety	Provisional literature extension
Individual	Agency	Provisional literature extension

Source labels are retained to preserve traceability, including "Longitivity" and "Social Engineering". The latter has divergent meanings in the source materials, which are documented in its QualCoder

memo. Coding distinguishes policy requirements, stated concerns, possible effects and observed effects. Each new code or category receives a memo recording its reference and source location, or its inductive origin and supporting corpus passages.

The complete exported codebook includes the live QualCoder memos, operational definitions, coding guidance and reference details. Regenerate it with `make coding-taxonomy-export`.



Threats to validity

- **Selection bias.** Popular projects may not represent the long tail; mitigated by explicit sampling criteria and a stratified sample.
- **Volatility.** Policies change; mitigated by a frozen and archived corpus.
- **Implicit sustainability.** Projects may address sustainability without naming it; mitigated by a broad keyword set and manual reading.
- **Coder subjectivity.** Mitigated by a double-coded sample and agreement metrics.
- **Policy versus practice.** Policies say little about actual behaviour; this is a stated limit of the study.

Expected contributions

- A map of how OSS communities frame sustainability when regulating GenAI.
- A coding grid of sustainability dimensions that can be reused.
- Recommendations for maintainers, for instance a template sustainability clause for AI policies.

References

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